| Program : **Diploma in Textile Technology** | |
|---|---|
| Course Code : **3061** | Course Title: **Fabric Manufacture I** |
| Semester : **3** | Credits: **3** |
| Course Category: **Program core** | |
| Periods per week: **3 (L:2, T:1, P:0)** | Periods per semester: **45** |

**Course Objectives:**

- To impart a basic knowledge on weaving preparatory process.
- Familiarize the students with basic mechanisms in power loom.
- To impart a basic knowledge on yarn numbering system, production and efficiency of weaving preparatory machines.

**Course Prerequisites:**

| Topic | Course code | Course Title | Semester |
|---|---|---|---|
| Knowledge of basic Mathematics | | Mathematics I | 1 |

**Course Outcomes:**

On completion of the course, the student will be able to:

| COn | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| CO1 | Understand the operations in warp and weft preparation process | 10 | Understanding |
| CO2 | Understand the warping ,sizing and drawing in and denting | 10 | Understanding |
| CO3 | Understand primary and secondary motions in a loom | 11 | Understanding |
| CO4 | Familiarize the yarn numbering, heald count, reed count and know the production and efficiency of weaving preparatory machines. | 12 | Applying |
| | Series Test | 2 | |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| **CO1** | 3 | | | | | | |
| **CO2** | 3 | | | | | | |
| **CO3** | 3 | | | | | | |
| **CO4** | | 3 | | | | | |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

<table>
<tr><th>Module Outcomes</th><th>Description</th><th>Duration (Hour)</th><th>Cognitive Level</th></tr>
<tr><td>CO1</td><td colspan="3">Understand the operations in warp and weft preparation process.</td></tr>
<tr><td>M1.01</td><td>Understand the warp winding operations in weaving.</td><td>5</td><td>Understanding</td></tr>
<tr><td>M1.02</td><td>Understand the weft winding operations in weaving.</td><td>5</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:<br>Objectives of Warp Winding - Understand the Working of Warp Winding Machines - Creel - types of tensioners - slub catchers - Broken thread stop motion - Full package stop motion - ribbon breakers - Package arm holders - traversing mechanism - features of Disc and Gate type tensioners - working of Electronic Yarn clearers - Warp Winding - Drum Winding and Precision winding - advantages of Electronic yarn clearers over ordinary slub catchers - Angle of wind - wind ratio - types of knots - splicing - Roto Coner - Auto Coner - defects in winding - causes & remedies Weft Winding Machines – Objectives - Schweiter high - speed automatic pirn winder - Bunching mechanism - Layer locking device - Pirn diameter control - defects in pirns, their causes and remedies.</td></tr>
<tr><td>CO2</td><td colspan="3">Understand the warping, sizing and drawing in and denting</td></tr>
<tr><td>M2.01</td><td>Know the warping operation in weaving preparatory process</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Understand the sizing operation in weaving preparatory</td><td>4</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Know the drawing in and denting operations in weaving</td><td>3</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test - I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:<br>Objectives of warping - beam warping and sectional warping - Ruti high speed warper - working sectional warping machine -Defects in warp beams, their causes and remedies - Objectives of sizing - ingredients used in the size paste and their functions- Steps involved in size paste preparation - multi cylinder sizing machine - functions of the various controls in a modern a sizing machine- size recipe for coarse, medium and fine cotton yarns-defects in sized beams - their causes and remedies - Objectives of drawing- in and denting – role of Healds and Reeds.</td></tr>
</table>

| | | | |
|---|---|---|---|
| **CO3** | **Understand primary and secondary motions in a loom** | | |
| M3.01 | Understand the shedding mechanism | 2 | Understanding |
| M3.02 | Understand the picking mechanism | 3 | Understanding |
| M3.03 | Know the beating up motion in a loom | 3 | Understanding |
| M3.04 | Understand the take up and let off motions | 3 | Understanding |

**Contents:**

Passage of material through a power loom- various parts and functions – Primary and secondary motions and role of each in weaving- Importance of Primary Motions- Principles of Shedding- shedding mechanisms -Tappet - Dobby - Jacquard - Different types of Sheds- working of Plain Tappet Shedding mechanism - Picking- objectives - Working of Cone Over Pick motion- Working of Under Pick motions- Discuss the beating operation- Explain the working of beating mechanism-Illustrate sley eccentricity - Secondary motions - take up motions - 7- wheel take up motion - continuous and intermittent take up motion- - Let off motions- working of a negative let off motion- working of Ropper let off motions.

| | | | |
|---|---|---|---|
| **CO4** | **Familiarize the yarn numbering, heald count, reed count and know the production and efficiency of weaving preparatory machines.** | | |
| M4.01 | Understand the yarn numbering system, Heald and Reed count | 6 | Understanding |
| M4.02 | Evaluate the production and efficiency of weaving preparatory machines | 6 | Applying |
| | Series Test - II | 1 | |

**Contents:**

Understand the different methods of Yarn Numbering Systems - -direct and indirect systems of yarn numbering - English, French, Denier, Tex - equations for calculating count in different Direct and

Indirect systems - resultant count of folded yarn - average count - beam count - heald and reed count. Speed, production and efficiency of warp winding machines - weft winding machines, warping machines - size percentage - production, speed and efficiency of sizing machines.

**Text / Reference:**

| T/R | Book Title/Author |
|---|---|
| R1 | The Mechanism of Weaving - Thomas W.Fox - Universal Publishing corp.,Mumbai-2 |
| R2 | Modern Preparation and Weaving Machinery - A.Ormerod - Woodhead publishing Ltd., England. |
| R3 | Weaving Mechanism Vol.I - N.N.Banerjee - Smt Tandra banerjee & Smt Apurba banerjee , West Bengal |
| R4 | Mechanism of Weaving Vol.I & II - Prof.J.C.Chakravarthi - Prof. .C.Chakravarthy West Bengal |
| R5 | Weaving Calculations - R.Sengupta- D.B.Taraporevala sons & co Ltd., Mumbai |

**Online resources:**

| Sl.No | Website Link |
|---|---|
| 1 | http://freetexbooks.blogspot.in/2011/10/handbook-of-weaving.html |
| 2 | http://www.nptel.ac.in/courses |
| 3 | http://textilelearner.blogspot.in |
| 4 | https://textilestudycenter.com |